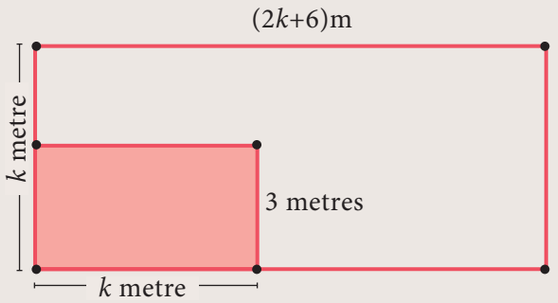
Activity 2

Consider a rectangular garden in front of a house, whose dimensions are $(2k+6)$ metre and k metre. A smaller rectangular portion of the garden of dimensions k metre and 3 metres is leveled. Find the area of the garden, not leveled.



Example 3.23 Find the zeroes of the quadratic expression $x^2 + 8x + 12$.

Solution Let $p(x) = x^2 + 8x + 12 = (x+2)(x+6)$

$p(-2) = 4 - 16 + 12 = 0$

$p(-6) = 36 - 48 + 12 = 0$

Therefore -2 and -6 are zeros of $p(x) = x^2 + 8x + 12$

Example 3.24 Write down the quadratic equation in general form for which sum and product of the roots are given below.

(i) 9, 14 (ii) $-\frac{7}{2}, \frac{5}{2}$ (iii) $-\frac{3}{5}, -\frac{1}{2}$

Solution (i) General form of the quadratic equation when the roots are given is

$$x^2 - (\text{sum of the roots})\,x + \text{product of the roots} = 0$$

$$x^2 - 9x + 14 = 0$$

(ii) $x^2 - \left(-\frac{7}{2}\right)x + \frac{5}{2} = 0 \Rightarrow 2x^2 + 7x + 5 = 0$

(iii) $x^2 - \left(-\frac{3}{5}\right)x + \left(-\frac{1}{2}\right) = 0 \Rightarrow \frac{10x^2 + 6x - 5}{10} = 0$

Therefore, $10x^2 + 6x - 5 = 0$.

Example 3.25 Find the sum and product of the roots for each of the following quadratic equations : (i) $x^2 + 8x - 65 = 0$ (ii) $2x^2 + 5x + 7 = 0$

(iii) $kx^2 - k^2x - 2k^3 = 0$

Solution Let α and β be the roots of the given quadratic equation

(i) $x^2 + 8x - 65 = 0$

$a = 1, \quad b = 8, \quad c = -65$

$$\alpha + \beta = -\frac{b}{a} = -8 \text{ and } \alpha\beta = \frac{c}{a} = -65$$

$\alpha + \beta = -8$; $\alpha\beta = -65$

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